GTM965500P

Test Report

24V Model

GlobTek Engineering

Generated: April 16, 2026

Test Summary

Project: GTM965500P | 24V Model | Total Test Items: 22

Status	OK / Resolved	For Review	Regression Req'd	In Progress	Has Issue	Invalid/ Incomplete Test	Unknown
Count	2	10	0	1	0	0	9

24V

Test Item	Status	Link
Input		
No Load Input Power	Marked for review	View →
Input Voltage	OK / Resolved	View →
Main Output		
Turn-On Delay	Marked for review	View →
Load Regulation	Marked for review	View →
Voltage Ripple	OK / Resolved	View →
Transient Response	Marked for review	View →
Startup Overshoot	Marked for review	View →
Hold-Up Time	Marked for review	View →
Remote-Sense (SNS)	Marked for review	View →
Output Voltage Fine Adjustment	Marked for review	View →
Protections		
Over-Power Protection (OPP)	Marked for review	View →
Short-Corcuit Protection (SCP)	Marked for review	View →

Changes Required

2 changes flagged across all tests.

Voltage	Test	Change Description
24V	Over-Power Protection (OPP) · *Vision314* · *2026-04-01*	COMPONENT | OLD VALUE | NEW VALUE -- | -- | -- C26 | 6.8nF | 5.6nF C27 | 3.3nF | 2.2nF
24V	Output Voltage Fine Adjustment · *Vision314* · *2026-04-02*	Req of Ra8, Ra9 and Ra11, Ra12 is 90.7kOhm Ra8, Ra11 = 88.7kOhm Ra9, Ra12 = 2kOhm Ra14 = 16kOhm Output range: 22.8V -> 25.2V

Tests Not Yet Run

10 tests have not yet been run.

24V

Test Item	Status	Link
Protections		
Over-Voltage Protection (OVP)	In Progress	View →
Environmental / Reliability		
Component Thermal Stress	Unknown	View →
Component Electrical Stress	Unknown	View →
Safety		
Dielectric Withstand (Hi-Pot)	Unknown	View →
Output Touch Current	Unknown	View →
Earth Leakage Current	Unknown	View →
EMC		
Conducted Emissions	Unknown	View →
Radiated Emissions	Unknown	View →
Electrostatic Discharge (ESD)	Unknown	View →
Electrical Fast Transient (EFT)	Unknown	View →

Detailed Test Results

24V

24V	No Load Input Power	OPEN	Marked for review

Specification / Procedure

Spec: < 500mW @ 240Vac (main-output disabled, i.e. remote-off low)

Solder RO pin to AUX GND to disable output without additional regulation board load

Use Yokogawa WT3000 with 20 second acquisition time (update rate)

Results

Vision314 · 2026-04-15 [LATEST]
570mW

Comments (1)

lauxtim · 2026-04-15 [LATEST]
See: https://github.com/GlobTek-Engineering/GTM965500P/issues/24#issuecomment-4255997812

24V	Turn-On Delay	OPEN	Marked for review

Specification / Procedure

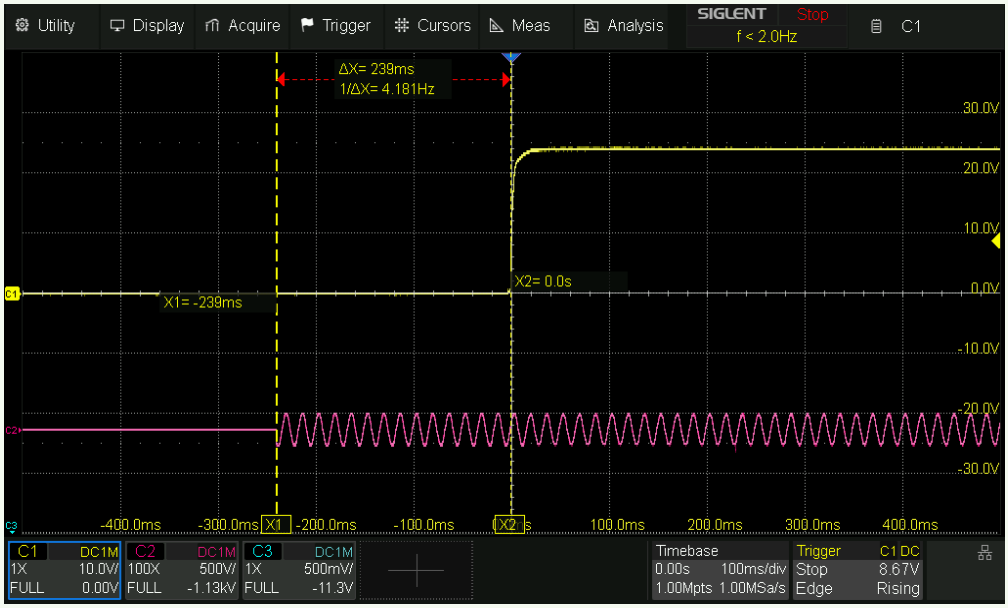
Spec: < 2 seconds @ 115Vac

Measure AC input voltage with differential probe, passive probe on DC output. Check the time between AC turn-on and DC turn-on, at full load.

Results

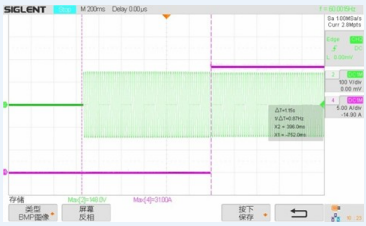
Vision314 · 2026-04-06

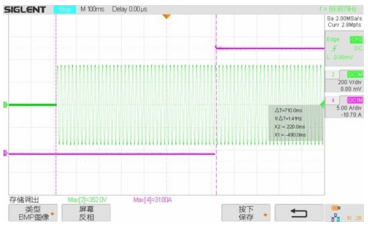
239ms @ 22.9A



Vision314 · 2026-04-16 [LATEST]

GTSZ

Condition	Spec (mS)	Reading (mS)	Picture	Result
100Vac, 22.916A	2000	1150		PASS

240Vac, 22.916A	2000	710		PASS

24V	Load Regulation	OPEN	Marked for review

Specification / Procedure

Spec: ± 1.0% (measured at output connector)

Check the stabilized DC output voltage at no-load (0%). Check the stabilized DC output voltage at full load (100%).

% Load regulation = [V(no_load) - V(full_load) / V(full_load)] * 100%

Measure at 90Vac
Measure at 264Vac

Results

Vision314 · 2026-04-06

Input	V (no load)	V (full load)	Load Regulation (%)
90Vac	24.036	24.033	0.01%
264Vac	24.036	24.032	0.02%

Vision314 · 2026-04-16 [LATEST]

GTSZ

Input Voltage	12V	Load (A)	Test Result
90	24.094	0	PASS
	24.02	5.73	PASS
	24.02	11.46	PASS
	24.01	17.19	PASS
	24.01	22.916	PASS
115	24.094	0	PASS
	24.02	5.73	PASS
	24.02	11.46	PASS
	24.01	17.19	PASS
	24.01	22.916	PASS
230	24.094	0	PASS
	24.02	5.73	PASS
	24.02	11.46	PASS
	24.01	17.19	PASS
	24.01	22.916	PASS
264	24.094	0	PASS
	24.02	5.73	PASS

	24.02	11.46	PASS
	24.01	17.19	PASS
	24.01	22.916	PASS

24V	Voltage Ripple	CLOSED	OK / Resolved

Specification / Procedure

Spec: < 1% pk-pk (measured with a 47µF low-ESR cap + 0.1µF ceramic cap, measured @ 20MHz BW)

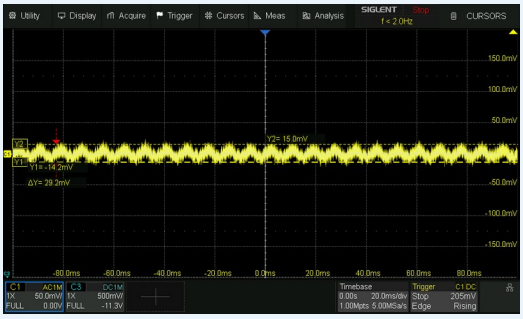
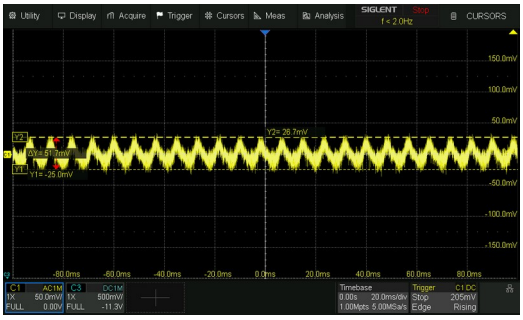
Prior to recording data, check whether 90Vac or 264Vac yields worse results, if at all. Record data using the worst case input voltage.

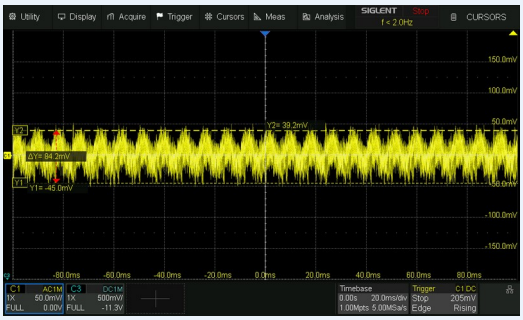
Using the ripple probe fixture and matching 1x probe, connect close to the output connectors on the PCB, measure ripple with oscilloscope. Ripple is inclusive of HF (100KHz) and LF (120Hz) components. Timebase should capture 120Hz component (e.g. 2-5ms/div)

Three conditions, adjust potentiometer accordingly:
(95% * V_out)
(100% * V_out)
(105% * V_out)

Results

Vision314 · 2026-04-06 [LATEST]
ALL PASS
1% ripple = 240mV

Output Voltage	Picture	Measured Ripple
95% Vout		29.2mV
100% Vout		51.7mV

105% Vout		84.2mV

24V	Transient Response	OPEN	Marked for review

Specification / Procedure

Spec: ± 5% deviation max. (with 50% load step), recovery to within ±1% within 1ms

Taken at nominal input voltage (e.g. 115Vac)

Measure with ripple probe fixture PCB, Measure at power supply PCB. Measured with AC coupling.

Step load + load dump:
0 -> 25%
25% -> 50%
50 -> 75%
75 -> 100%

8 measurements total

Results

Vision314 · 2026-04-06 [LATEST]

Load Step

Transition	Deviation	Recovery Time
0A → 5.7A	493mV	<1ms
5.7A → 11.5A	248mV	<1ms
11.5A → 17.2A	255mV	<1ms
17.2A → 22.9A	303mV	<1ms

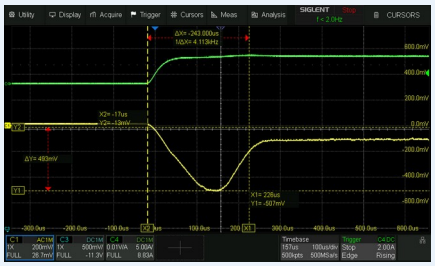
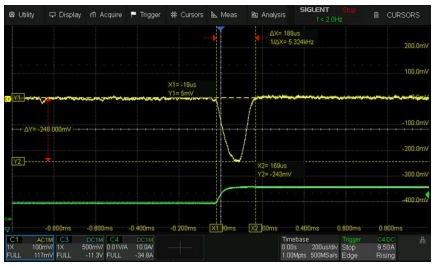
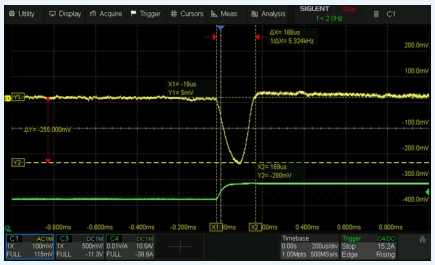
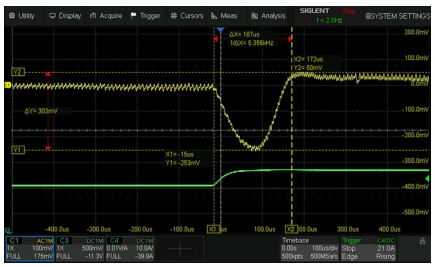
Load Dump

Transition	Deviation	Recovery Time
0A → 5.7A	220mv	<1ms
5.7A → 11.5A	182mV	<1ms
11.5A → 17.2A	181.7mV	<1ms
17.2A → 22.9A	207.5mV	<1ms

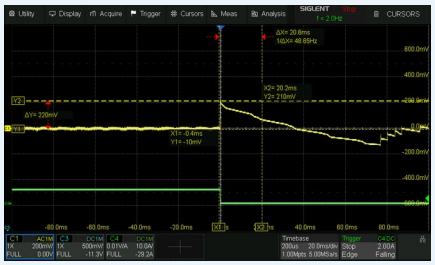
1% = 240mV

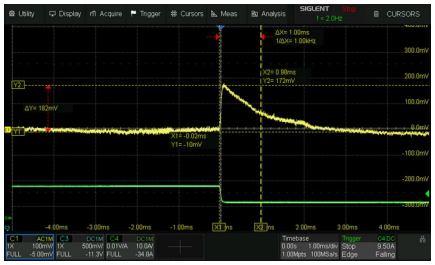
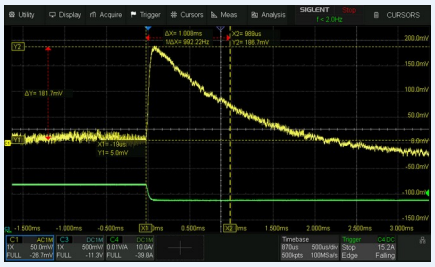
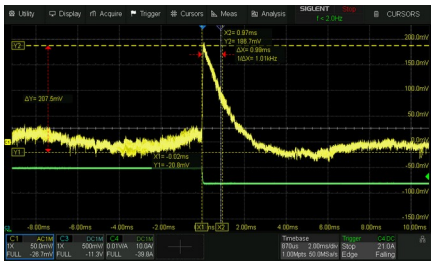
Load Step

Load Step	Picture	Deviation	Recovery Time

0A → 5.7A		493mV	<1ms
5.7A → 11.5A		248mV	<1ms
11.5A → 17.2A		255mV	<1ms
17.2A → 22.9A		303mV	<1ms

Load Dump

Load Dump	Picture	Deviation	Recovery Time
0A → 5.7A		220mv	<1ms

5.7A → 11.5A		182mV	<1ms
11.5A → 17.2A		181.7mV	<1ms
17.2A → 22.9A		207.5mV	<1ms

Comments (1)

lauxtim · 2026-04-14 [LATEST]

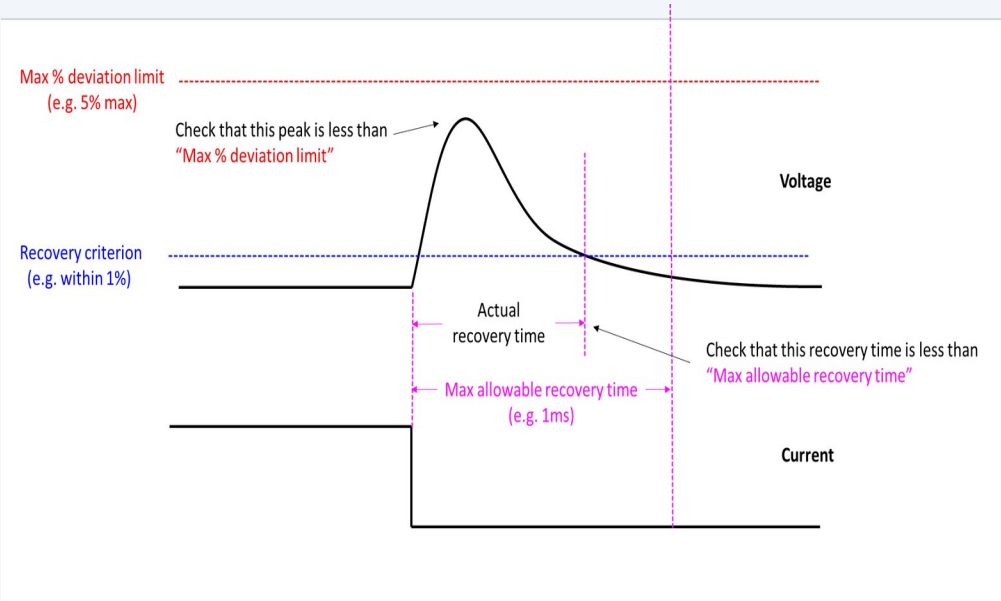
In reviewing the transient response for 12, 24V, and 54V, many times the peak deviation does not ever exceed the 1% recovery threshold, so the recovery time should probably be marked as "N/A" or "Dev. < 1%" for those items. (Cannot quantify how long it takes to recovery if it never deviated outside of our bounds to start with.) Above, all of the load steps exceed 1% deviation (240mV), so then we need to assess whether the peak deviation has exceeded the 5% max or not. From inspection, seems like most are in the 200-300us range. Writing <1ms is OK as a pass/fail. For the future, you *could* record the actual recovery time.

No need to re-test, just update those which never deviated outside of 1% to "Dev. < 1%" or similar.

As a note:

I'd probably set the cursors as follows:

Y(1): At peak of measured waveform (measured max deviation)
Y(2): At the 1% recovery threshold
X(1): At t=0 (the moment the load step is applied)
X(2): At the moment the voltage recovers to less than Y(2), the recovery threshold (measured recovery time)



It may make sense (in general, all types of tests), to present data with the specified limits presented in line. This way, you can easily see (at a glance) whether the spec is met or not, no need to scroll up or remember what the spec is.

Load Dump Condition	Picture	Max Deviation	Max Deviation (Spec)	Recovery Time	Recovery Time (Spec)

24V	Startup Overshoot	OPEN	Marked for review

Specification / Procedure

Spec: <5%

Measure with oscilloscope in DC coupling.

Measure at 90Vac and 264Vac, at:
No load
Full load

4 measurements total

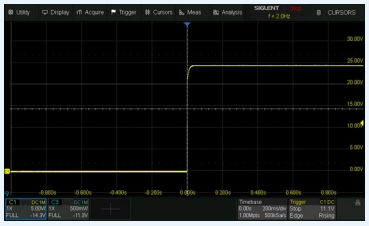
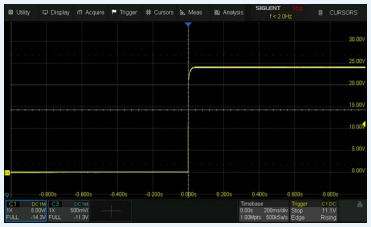
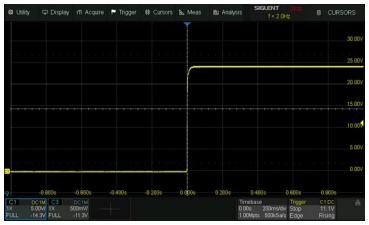
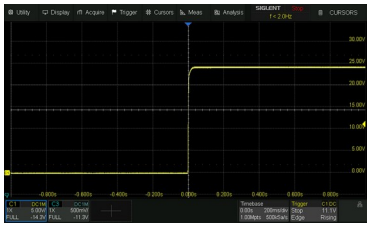
Results

Vision314 · 2026-04-06 [LATEST]

Output Voltage Overshoot

Condition	90 Vac	264 Vac
No Load	0mV	0mV
Full Load	0mV	0 mV

Output Voltage Overshoot

Condition	90 Vac	264 Vac
No Load		
Full Load		

24V	Hold-Up Time	OPEN	Marked for review

Specification / Procedure

Spec: 20ms typ. (100% load), 20ms min. (95% load), 50ms typ. (50% load)

Attach differential probe to AC input. Attach passive probe to output. Set load according to the above. Measure the time between AC turn-off and DC output falling to approx. 95% * nominal V_out.

Results

Vision314 · 2026-04-02 [LATEST]

Input Voltage	Output Voltage	Hold Up Time (ms) 100% Load (22.9 A)	Hold Up Time (ms) 50% Load (11.5 A)
90Vac	22.8	28.3	56.3
	24	27.5	52.3
	25.7	25.5	50.7
264Vac	22.8	28.9	56.3
	24	27.3	52.3
	25.7	25.7	50.1

24V	Remote-Sense (SNS)	OPEN	Marked for review

Specification / Procedure

Spec: Up to 500mV total cable-drop compensation, reverse and short circuit protected. Connect SNS+/SNS- leads to point-of-load with twisted pair wire. (Note 1)

Attach sufficiently long wires between the output and the load, which causes a voltage drop exceeding 250mV per wire @ 85% load current. (Rationale for 85%: When remote sense is used, output voltage and output power goes up, when tested with a constant current load. We do not specify operation at P_out>100% rating.)

For instance, the DCR of 1.5m of 12AWG wire is 0.0075ohm. For the 12V model, this yields a drop of 292mV per wire, or 584mV total. Different cable lengths shall be used for other voltage models.

Part A:

- Run at nominal line voltage (e.g. 115Vac)
- Increase the set-point voltage to V_nom + 5% using the potentiometer.

1. Leave the sense leads disconnected from the load. Measure the voltage at the power supply PCB @ 85% load.

2. Connect the twisted-pair sense leads at the point-of-load (at the E-load terminals).

3. Measure the voltage at the power supply PCB again @ 85% load.

The voltage at the PCB should have increased by at least 500mV.

Part B:

1. Detach the SNS leads, reverse the leads (connect SNS+ to -load terminal and SNS- to +load terminal). Check that there is no damage during and after.

Part C:

1. Detach the SNS leads and connect them to each other (short SNS+ to SNS-, alone). Check that there is no damage during and after.

Results

Vision314 · 2026-04-13 [LATEST]
Previous issues were caused by broken/missing/incorrect TVS diodes for ZD6 and ZD7. Must be 54V zener ("RE" marking) for all voltage models.

85% load = 19.5A

Part A

Step #	Voltage @ Supply	Voltage @ Load
1	25.2 V	24.25 V
3	25.8 V	24.9 V

Part B - No damage

Voltage @ Supply	Voltage @ Load
26.45 V	25.5 V

Part C - No damage

Voltage @ Supply	Voltage @ Load
26.3 V	25.5 V

Comments (3)

Vision314 · 2026-04-07

Part A

12AWG	~6ft	Rwire = 0.013 Ohm

85% load = 19.5A
1. 25.3 V
2. ..
3. 25.67 V

Part B

1. When I connected the SNS- to +load, there was a spark between those two terminals and the supply OVP'd. On startup the output voltage goes to ~40V and OVP's.

Part C
1. ...

Vision314 · 2026-04-08

The ZD7 TVS diode was removed previously for other rework troubleshooting

When the SNS+ connection touched the negative output terminal, this create some large voltage spike that damaged RJ2
RJ2 Failure Mode: open

Replacing RJ2 with another 2.2Ohm resistor fixed the supply

Solution: don't remove TVS diodes, we need them...

I will finish SNS testing shortly

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lauxtim · 2026-04-15 [LATEST]

> See: https://github.com/GlobTek-Engineering/GTM965500P/issues/55#issuecomment-4256097729

24V	Output Voltage Fine Adjustment	OPEN	Marked for review

Specification / Procedure

Spec: Via potentiometer, ±5% adjustment range (Note 1)

22.8V -> 25.2V

Run at nominal line voltage (e.g. 115Vac)

Sense leads (SNS) disconnected.

Adjust the potentiometer to minimum and maximum positions. Record the minimum and maximum output voltage, measured at the PCB.

Changes Required

Vision314 · 2026-04-02 [LATEST]
Req of Ra8, Ra9 and Ra11, Ra12 is 90.7kOhm

Ra8, Ra11 = 88.7kOhm
Ra9, Ra12 = 2kOhm
Ra14 = 16kOhm

Output range: 22.8V -> 25.2V

24V	Over-Power Protection (OPP)	CLOSED	Marked for review

Specification / Procedure

Spec: 105-150%, non-latching with auto-recovery

Spec update: Change from "Over-Current Protection" to "Over-Power Protection"

Run at 90Vac and 264Vac.

The specified range applies only for V_out = V_out(nom). (e.g. "12V" @ 12.00V, "24V" @ 24.00V, etc.), voltage measured at the PCB. The nominal design point should be 130 - 135%.

Reference OPP data should also be collected at V_out(-5%) and V_out(+5% + 500mV), for reference although not specified. At V_out(+5% + 500mV), at least 100% output power should be obtainable.

FRANKIE TESTING NOTES:
4/1/2026
Noting current values for regulation board. In order to test Vout max and min (22.8 V -> 25.7V) the following regulation board components were used and give a actual output voltage range of 21.4 V -> 26.5V

Ra8, Ra11 = 84.5kOhm
Ra9, Ra12 = 100Ohm
Ra10, Ra13 = 10kOhm

Ra14 = 7.5kOhm

Changes Required

Vision314 · 2026-04-01 [LATEST]

COMPONENT	OLD VALUE	NEW VALUE
C26	6.8nF	5.6nF
C27	3.3nF	2.2nF

Results

Vision314 · 2026-04-14 [LATEST]

Input Voltage	Output Voltage	OPP Point (A)	OPP Point (%)	PG Time (ms)
90Vac	22.8	33	144%	10
	24	32	140%	10
	25.7	24	112%	10
264Vac	22.8	38	166%	10
	24	32	140%	10
	25.7	24	112%	10

Comments (1)

Vision314 · 2026-04-01 [LATEST]
Trying the same values as the 12V:

COMPONENT	OLD VALUE	NEW VALUE
C26	6.8nF	6.8nF
C27	3.3nF	2.2nF

Input Voltage	Output Voltage	OPP Point (A)	OPP Point (%)	PG Time (ms)
90Vac	22.8	34	148%	0
	24	32	140%	9.5
	25.7	27	126%	9.5
264Vac	22.8	43	188%	9.5
	24	36	157%	9.5
	25.7	27	126%	9.5

FRANKIE TESTING NOTES:
4/1/26
Noting current values for regulation board. For testing between -5% to +(5% + 500mV) => 22.8 V -> 25.7V

21.4 V -> 26.5V

Ra8, Ra11 = 84.5kOhm
Ra9, Ra12 = 100Ohm
Ra10, Ra13 = 10kOhm

Ra14 = 7.5kOhm

24V	Short-Corcuit Protection (SCP)	OPEN	Marked for review

Results

Vision314 · 2026-04-15 [LATEST]
PASS

24V	Input Voltage	CLOSED	OK / Resolved

Specification / Procedure

Spec: 90 - 264Vac (Nameplate: 100 - 240Vac)

Confirm that full rated power can be drawn over the entire rated input voltage range. Test full load @ 90Vac, 115Vac, 240Vac, 264Vac

Results

Vision314 · 2026-04-16 [LATEST]

GTSZ

Input Voltage	Output Voltage	Load (A)	Test Result
90	24.094	0	PASS
90	24.01	22.916	PASS
120	24.094	0	PASS
120	24.01	22.916	PASS
240	24.094	0	PASS
240	24.01	22.916	PASS
264	24.094	0	PASS
264	24.01	22.916	PASS

Comments (1)

Vision314 · 2026-04-02 [LATEST]
https://github.com/GlobTek-Engineering/GTM965500P/issues/68#issuecomment-4172278539

Untested — Specifications

24V

24V	Over-Voltage Protection (OVP)	OPEN	In Progress
24V	Component Thermal Stress	OPEN	Unknown
24V	Dielectric Withstand (Hi-Pot)	OPEN	Unknown
24V	Output Touch Current	OPEN	Unknown
24V	Earth Leakage Current	OPEN	Unknown
24V	Conducted Emissions	OPEN	Unknown
24V	Radiated Emissions	OPEN	Unknown
24V	Electrostatic Discharge (ESD)	OPEN	Unknown
24V	Electrical Fast Transient (EFT)	OPEN	Unknown
24V	Component Electrical Stress	OPEN	Unknown